Re-analysis Summary Report for Peer-evaluation

Paper ID: **Hurst_EvoHumanBehavior_2017_yypJ**

Re-analyst's ID: **ysqya**

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: https://forms.gle/EAREaC6JjN2PWFfR6

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:
... those with a faster life strategy report greater levels ... of psychopathology (p. 1.)

The corresponding article

Here you can find the original article: https://osf.io/tqmbe/?view_only=c874fcf097644e82a320679d4641163a and the original data: https://osf.io/vhmgc/?view_only=5c6bedb36b2549a88ac137d6c746bcb8

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:
I calculated the sum of all the miniK items, all the HKSS items, and all the DSM-5 items, respectively. Because I planned to conduct four analyses at the same time, the p-value was corrected at .0125. To test the hypothesis that people with a faster life strategy report greater levels of psychopathology, I first conducted partial correlation between miniK and DSM-5 with age controlled—this followed exactly what the authors did in the article. The correlation between miniK and DSM-5 was r = -.51, p = < .001. Second, I tested a regression model with DSM-5 predicted by miniK while age being controlled; the estimate for miniK was -.64, p < .001, = -.50. Therefore, people with lower scores on miniK reported more symptoms of psychopathology. Then, I did the same analyses for HKSS and the correlation between HKSS and DSM-5 was r = -.41, p = < .001; the estimate for HKSS in regression was -.41, p < .001, = -.39. Therefore, people with lower scores on HKSS reported more symptoms of psychopathology. Overall, the hypothesis was supported.

The analyst's conclusion: Overall, the hypothesis that people with a faster life strategy report greater levels of psychopathology was supported.

The analyst's categorization of the result:
The results show evidence for the relationship/effect as described in the claim provided in your task

In Task 2, the analysts received the following instruction to conduct the analysis:
You should report the Mini-K measure instead of HKSS for life history strategy.

Here, the analyst reported the following steps and results of the analysis:
I summed up the scores of the Mini-K items and the DSM-5 items, respectively. Because I planned to conduct two analyses at the same time, the p-value was corrected at .025. To test the hypothesis that people with a faster life strategy report greater level of psychopathology, I conducted partial correlation between Mini-K and DSM-5 with age controlled—same analysis as what the authors did in the article. The correlation between Mini-K and DSM-5 was $r = -.51$, $p = < .001$. Therefore, people with lower scores on Mini-K reported more symptoms of psychopathology. Thus, the hypothesis (people with a faster life strategy report greater level of psychopathology) was supported and a successful replication was achieved.

Analysis language/software/code (optional to check):
Task 1: R
Task 2: R
Analysis codes: https://osf.io/w5ea6/